

Recombination Rate Coefficients in EXHALE: H I, He I, He II, and the He I 2³S Triplet, with Benchmarks and the Status of Newer Data

Contents

1. Overview	1
2. What EXHALE uses	1
3. H ⁺ → H: case-A vs. case-B, not data vintage	2
4. Helium recombination	3
5. Is there newer data?	3
6. Reproducing this comparison	5

1. Overview

This memo documents how EXHALE treats recombination onto H I, He I, He II, and the metastable He I 2³S level (the lower level of the He 10830 line), benchmarks the adopted coefficients against other published fits, and reviews whether more recent data warrant an update. The companion to this is the photoionization cross-section memo (photoion_cross_sections.tex); together they cover the two microphysical inputs that set the H/He ionization balance and the He 2³S population.

The two headline conclusions are: (i) EXHALE's H/He *case-B* coefficients (Hui & Gnedin 1997) agree with the Draine (2011) case-B fits to $\sim 2\%$ over $5 \times 10^3 - 2 \times 10^4$ K — they are not outdated; the larger differences seen against some escape codes are the *case-A vs. case-B* modeling choice, not data vintage. (ii) EXHALE's He 2³S triplet recombination (Benjamin, Skillman & Smits 1999, as parametrized by Oklopčić & Hirata 2018) is *identical* to that adopted by the most recent multi-species escape model (Taylor et al. 2025), so it reflects current escape-community practice.

2. What EXHALE uses

All H/He recombination coefficients are defined in `src/modules/radiation/Cool_coeff.f90` and assembled in `ionization_equilibrium.f90`. Units are $\text{cm}^3 \text{s}^{-1}$.

H II → H I and He III → He II (Hui & Gnedin 1997, case B). With $\lambda_i = 2T_i/T$ and T_i the ionization temperature (157807, 285335, 631515 K for H I, He I, He II),

$$\alpha_{\text{H II}}^B(T) = 2.753 \times 10^{-14} \frac{\lambda_{\text{H}}^{1.5}}{[1 + (\lambda_{\text{H}}/2.74)^{0.407}]^{2.242}}, \quad (1)$$

$$\alpha_{\text{He III}}^B(T) = 2 \times 2.753 \times 10^{-14} \frac{\lambda_{\text{He}^+}^{1.5}}{[1 + (\lambda_{\text{He}^+}/2.74)^{0.407}]^{2.242}}, \quad (2)$$

the second being the hydrogenic ($Z = 2$) scaling of the first. Case B (the “on-the-spot” approximation) excludes direct recombination to the ground state, whose re-emitted photon is assumed re-absorbed locally.

He II → He I (two regimes). When the He 2³S network is *off*, EXHALE uses the Hui & Gnedin (1997) total case-B coefficient $\alpha_{\text{He II}}^B = 1.26 \times 10^{-14} \lambda_{\text{He}}^{0.75}$. When the network is *on*, the He II → He I recombination is split into singlet and triplet channels (ionization_equilibrium.f90; the total case-B value is overwritten), following Benjamin, Skillman & Smits (1999) as parametrized by Oklopčić & Hirata (2018):

$$\alpha_1(T) = 1.54 \times 10^{-13} (T/10^4)^{-0.486} \quad (\text{singlet, } 1^1S), \quad (3)$$

$$\alpha_3(T) = 2.10 \times 10^{-13} (T/10^4)^{-0.778} \quad (\text{triplet, } 2^3S). \quad (4)$$

These are *effective* (cascade-summed) coefficients: α_3 is the total rate at which recombinations populate the metastable 2³S level via direct capture plus radiative cascades through the higher triplet terms. The triplet captures the majority of He recombinations, $\alpha_3/(\alpha_1 + \alpha_3) \simeq 0.53\text{--}0.63$ over $5 \times 10^3\text{--}2 \times 10^4$ K (Table 2).

Metals use the modern Badnell radiative + dielectronic recombination (RR+DR), described in the cooling memo, and are not revisited here.

3. H⁺ → H: case-A vs. case-B, not data vintage

Figure 1 and Table 1 compare EXHALE’s case-B $\alpha_{\text{H II}}$ with the Draine (2011) case-A/B fits and with the total (case-A) coefficient adopted by the recent escape model of Taylor et al. (2025, their Storey & Hummer 1995 value). EXHALE tracks the Draine *case-B* curve to $\sim 2\%$ across the whole range. The Taylor+2025 value is $\sim 40\text{--}90\%$ higher, but that is because it is a *case-A* (total) rate: codes that transport the diffuse recombination continuum explicitly use case A, whereas EXHALE’s on-the-spot scheme correctly uses case B. The underlying data agree; the difference is the radiative-transfer treatment.

Table 1: H⁺ → H recombination coefficient ($10^{-13} \text{ cm}^3 \text{ s}^{-1}$).

T [K]	EXHALE (B)	Draine (B)	Draine (A)	Taylor+25 (A)
5000	4.543	4.428	6.733	6.608
10000	2.592	2.540	4.130	4.241
15000	1.837	1.818	3.087	3.271
20000	1.428	1.428	2.505	2.721

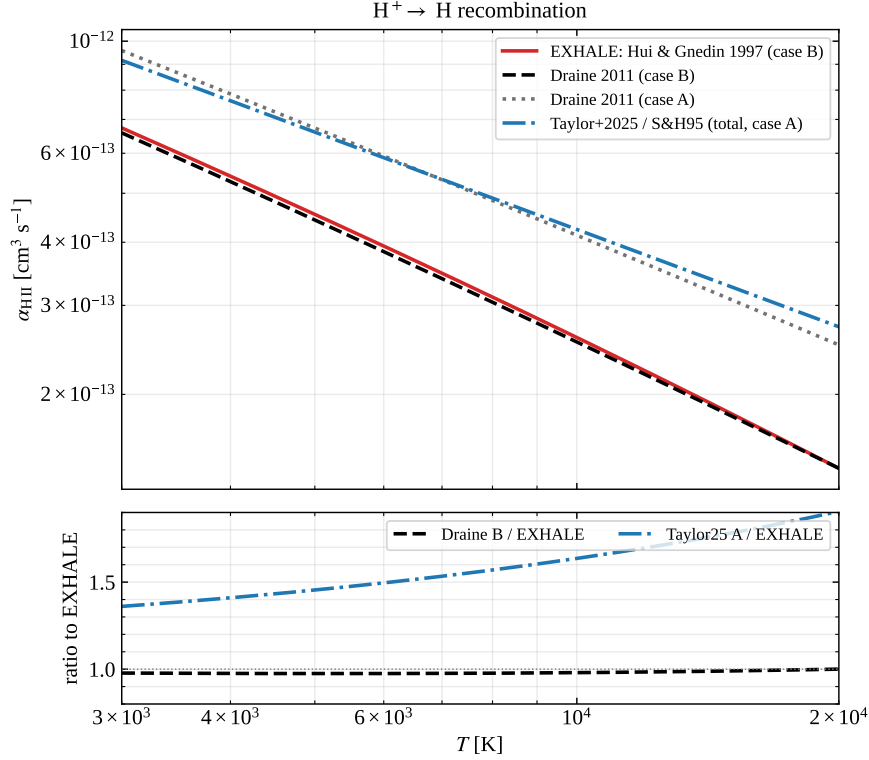


Figure 1: $H^+ \rightarrow H$ recombination. EXHALE (Hui & Gnedin 1997, case B; red) overlaps the Draine (2011) case-B fit to $\sim 2\%$ (bottom panel); the case-A (total) curves of Draine and Taylor+2025 are $\sim 40\text{--}90\%$ higher. The offset is the case-A vs. case-B choice, not a data update.

4. Helium recombination

The left panel of Fig. 2 repeats the case-A/B story for $He^{++} \rightarrow He^+$: EXHALE (case B) matches the hydrogenic Draine case-B to $\sim 2\%$, while the Verner & Ferland (1996) total used by Taylor+2025 is $\sim 45\%$ higher (case A). The right panel shows $He^+ \rightarrow He$: the Hui & Gnedin total case-B, and the Benjamin+1999 singlet/triplet split that EXHALE uses when the 2^3S network is active. Table 2 lists the values.

Two points are worth noting. (i) The triplet recombination α_3 (the quantity that feeds He 10830) is *identical* in EXHALE and in the latest escape models — both adopt Benjamin+1999. (ii) The Benjamin+1999 effective total $\alpha_1 + \alpha_3$ is $\sim 40\%$ larger than the Hui & Gnedin (1997) total case-B (e.g. 3.64 vs. $2.62 \times 10^{-13} \text{ cm}^3 \text{ s}^{-1}$ at 10^4 K); the two come from different methods (effective level-resolved cascade sums vs. a recombination-spectrum case-B total). When the 2^3S network is on, EXHALE consistently uses the Benjamin split for both channels, so the He ionization balance and the metastable source term are computed from the same data set.

5. Is there newer data?

H I and He II (hydrogenic). Essentially solved. Hui & Gnedin (1997), Storey & Hummer (1995), Verner & Ferland (1996), and Badnell (2006) agree to $\sim 1\text{--}2\%$ (Fig. 1). No update is warranted; the only meaningful choice is case A vs.

Table 2: He recombination coefficients ($10^{-13} \text{ cm}^3 \text{ s}^{-1}$). Left: $\text{He}^{++} \rightarrow \text{He}^+$. Right: $\text{He}^+ \rightarrow \text{He}$ — Hui & Gnedin total (case B) and the Benjamin+1999 singlet α_1 , triplet α_3 , and triplet fraction.

T [K]	$\text{He}^{++} \rightarrow \text{He}^+$			$\text{He}^+ \rightarrow \text{He}$			
	EXHALE(B)	Draine(B)	Taylor25(A)	HG97 tot	α_1	α_3	$\alpha_3/(\alpha_1+\alpha_3)$
5000	25.61	25.35	34.51	4.400	2.157	3.601	0.625
10000	15.45	15.13	22.61	2.616	1.540	2.100	0.577
15000	11.37	11.09	17.66	1.930	1.265	1.532	0.548
20000	9.09	8.86	14.82	1.556	1.100	1.225	0.527

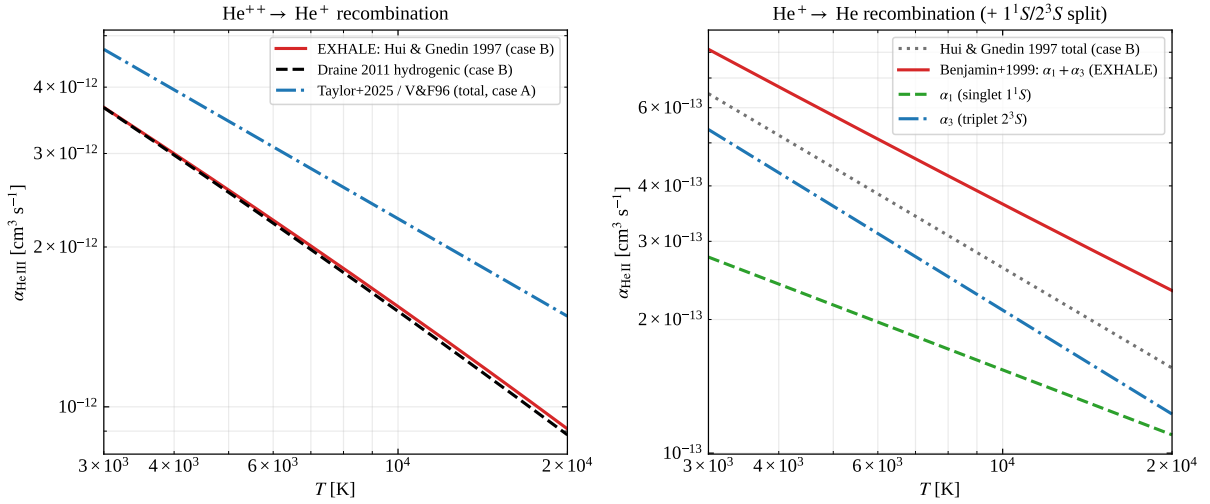


Figure 2: Helium recombination. **Left:** $\text{He}^{++} \rightarrow \text{He}^+$ — EXHALE case B (red) $\approx$ Draine hydrogenic case B (black); Verner & Ferland 1996 / Taylor+2025 total (case A) is $\sim 45\%$ higher. **Right:** $\text{He}^+ \rightarrow \text{He}$ — Hui & Gnedin total case B (dotted) and the Benjamin+1999 singlet/triplet split ($\alpha_1 + \alpha_3$, α_1 , α_3) used by EXHALE when the 2^3S network is active.

B (§3.).

He I total and the 2^3S triplet — no update available. The newer He I references often cited here — **Porter, Ferland, Storey & Detisch (2012, MNRAS 425, L28; 2013 erratum MNRAS 433, L89)** (the Cloudy default) and **Del Zanna & Storey (2022)** — are *emissivity* papers. They improve the He I *line cascade* / *line-ratio* predictions, but they do *not* provide, and do not change, the quantity EXHALE needs: the *total* (effective) recombination into the triplet ladder, α_3 , which all funnels to the metastable 2^3S . That total is set by (case-B total recombination) $\times$ (triplet spin fraction $3/4$) — ultimately by the photoionization cross sections (Milne) and spin statistics — and is *robust* across modern treatments:

- EXHALE (Benjamin+1999): $\alpha_3 = 2.10 \times 10^{-13} T_4^{-0.778}$, $\alpha_1 = 1.54 \times 10^{-13} T_4^{-0.486} \text{ cm}^3 \text{ s}^{-1}$;
- 2025 thermosphere model (arXiv:2509.14499; from Draine 2011 §14.3): $\alpha_3 = \frac{3}{4} (2.72 \times 10^{-13} T_4^{-0.789}) = 2.04 \times$

$10^{-13} T_4^{-0.789}$, with the *identical* singlet $\alpha_1 = 1.54 \times 10^{-13} T_4^{-0.486} \text{ cm}^3 \text{ s}^{-1}$.

α_3 agrees to $\sim 2\text{--}3\%$ and α_1 is identical. The most recent multi-species escape model (Taylor et al. 2025) likewise adopts Benjamin+1999 for the triplet. Attempting to extract a different value from the local Cloudy install confirmed the point: its `he_iso_recomb.dat` stores per-level direct radiative recombination, and summing all triplet ($S=3$) captures *is* the effective $\alpha(2^3S)$ in case B, but the file carries no usable level $\rightarrow (n, \ell, S)$ map for its 1642 He levels (only 111 are resolved in `helike_energies.dat`) and a trial sum overshoots the known case-A total by $\sim 2\times$ — i.e. there is no clean drop-in Cloudy number either.

Recommendation. **Leave the recombination as it is.** EXHALE’s Benjamin+1999 α_1, α_3 already match the latest escape-modeling practice and the underlying atomic physics to $\lesssim 3\%$; the Porter/Del Zanna “updates” are line-emissivity improvements that do not alter the total 2^3S recombination. The H/He case-B rates likewise need no change. The single worthwhile atomic-data update in this round is the *photoionization* of the He I ground state (Verner+1996, now the EXHALE default; see docs/Update_EXHALE §22), which shifts the He 10830 depth by $\sim 3\%$.

6. Reproducing this comparison

```
cd ATES/EXHALE/docs
python3 compare_recombination.py
# -> figures/rec_H.pdf, figures/rec_He.pdf
```

The script ports the EXHALE coefficients from `Cool_coeff.f90` verbatim and codes the Draine (2011), Verner & Ferland (1996) / Taylor+2025, and Benjamin+1999 reference fits.